Post-quantum Algorithms

Mihai Prunescu *

Abstract

Various post-quantum algorithms and protocols will be exposed here.

1 Introduction

In April 2026 Google announced that they broke a key-exchange protocol based on ellipticvc curves using a quantum computer of only 10^5 qubits. Possibly, this announcement will accelerate the implementation of post-quantum security faster than previously expected. In this pages, I will present various cryptosystems and protocols. If the progress will take place as expected, in only few years algorithms like Diffie-Hellman, RSA, Elgamal or Goldwasser-Micali will become only legacy. It is weird, but it is what it is.

2 Oded Regev's Cryptosystem

This Cryptosystem is the basis of Kyber, one of the winner of the NIST Post-Quantum Competition. See [1].

We let n be the security parameter of the cryptosystem. Our cryptosystem is parameterized by two integers m, p and a probability distribution χ on $\mathbb{Z}_p$. A setting of these pa- rameters that guarantees both security and correctness is the following. Choose $m = 5n$ and set $p \geq 2$ to be some prime number between n^2 and $2n^2$. The probability distri- bution χ is taken to be $\Psi_{\alpha(n)}$ for $\alpha(n) = o(1/\sqrt{n \log n})$, i. e., $\alpha(n)$ is such that

$$\lim_{n\to\infty} \alpha(n)/\sqrt{n \log n} = 0.$$

For example, we can choose $\alpha(n) = 1/(\sqrt{n} \log n)$. In the following description, all additions are performed in $\mathbb{Z}_p$, i.e., modulo p.

Private key. Choose $\vec{s} \in \mathbb{Z}_p^n$ uniformly at random.

Public key. For $i = 1, \dots, m$ choose m vectors $\vec{a}_1 \dots, \vec{a}_m \in \mathbb{Z}_p^m$ independently from the uniform distribution. Also choose elements $e_1, \dots, e_m \in \mathbb{Z}_p$ independently according to χ. The public key is given by $(\vec{a}_i, b_i)_{i=1}^m$, where $b_i = \langle \vec{s}, \vec{a}_i \rangle + e_i$.

Encryption. In order to encrypt a bit $x \in \{0, 1\}$, we choose a random set $S \subseteq \{1, \dots, m\}$. The encryption is:

$$\left(\sum_{i\in S} \vec{a}_i \quad , \quad x \cdot \left\lfloor \frac{p}{2} \right\rfloor + \sum_{i \in S} b_i \right).$$

*Faculty of Mathematics and Informatics, University of Bucharest, Romania and Simion Stoilow Institute of Mathematics of the Romanian Academy, Research unit 5, P. O. Box 1-764, RO-014700 Bucharest, Romania. mihai.prunescu@gmail.com

Decryption. The decryption of a pair $(\vec{a}, b)$ is $x = 0$ if $b - \langle \vec{a}, \vec{s} \rangle$ is closer to 0 than to $\lfloor \frac{p}{2} \rfloor$ on the circle modulo p. Otherwise, the decryption is $x = 1$.

The main observation is, that this decryption works only with big probability, but not with probability 1, so one has to combine the method with error-correcting codes. Another observation: the public key for Kyber is around 1.5 MB large.

References

[1] **Oded Regev**: *On lattices, learning with errors, random linear codes, and cryptography.* STOC05: Symposium on Theory of Computing, 2005.

https://dl.acm.org/doi/10.1145/1060590.1060603